

Multiple Choice

1. **Answer:** C

Options: A) 54.91 MHz; B) 239.2 MHz; C) 345.0 MHz; D) 2167 MHz

Note: Correct option: C - 345.0 MHz

2. **Answer:** D

Options: A) The most common oxidation state for the lanthanide elements is +3.; B) Lanthanide complexes often have high coordination numbers (> 6).; C) All of the lanthanide elements react with aqueous acid to liberate hydrogen.; D) The atomic radii of the lanthanide elements increase across the period from La to Lu.

Note: Correct option: D - The atomic radii of the lanthanide elements increase across the period from La to Lu.

3. **Answer:** D

Options: A) Both ^{35}Cl and ^{37}Cl have $I = 0$.; B) The hydrogen atom undergoes rapid intermolecular exchange.; C) The molecule is not rigid.; D) Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.

Note: Correct option: D - Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.

4. **Answer:** D

Options: A) The process is exothermic.; B) The process does not involve any work.; C) The entropy of the system increases.; D) The total entropy of the system plus surroundings increases.

Note: Correct option: D - The total entropy of the system plus surroundings increases.

5. **Answer:** D

Options: A) 10.8×10^{-5} ; B) 4.11×10^{-5} ; C) 3.43×10^{-5} ; D) 1.71×10^{-5}

Note: Correct option : D - 1.71×10^{-5}

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Tl'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'reduced mass'.

Long Form Response

11. **Answer:** Cr₃+. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** B₁₂ icosahedra. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key